

Memory Card Matching Game Enhancement Exercise

Objective:

Enhance your basic Memory Card Matching Game by adding new features and improvements. This exercise will help you practice and solidify your skills in HTML, CSS, and JavaScript.

Instructions:

1. Change the Grid Layout:

- **Task:** Modify the game board from a fixed 4x4 grid to support various grid sizes (e.g., 4x4, 6x6).
- **Hints:**
 - Update the HTML and CSS grid properties (e.g., grid-template-columns) to accommodate a dynamic number of cells.
 - Adjust the JavaScript logic to handle a varying number of cards based on the selected grid size.

2. Add a Move Counter and Timer:

- **Task:** Implement a scoreboard that tracks and displays the number of moves and the time taken to complete the game.
- **Hints:**
 - Create HTML elements for the move counter and a timer display above or below the game board.
 - Use JavaScript variables to keep track of the moves and elapsed time, updating the display as the game progresses.
 - Consider using setInterval to update the timer every second.

3. Enhance Card Flipping Effects:

- **Task:** Improve the visual appeal of the card flipping effect.
- **Hints:**
 - Use CSS transitions and 3D transforms (e.g., transform-style: preserve-3d) to create a smoother flip animation.
 - Divide each card into a “front” and “back” element so that the flip effect reveals the card’s face.
 - Add hover effects to indicate that cards are clickable.

4. Implement “Reset Game” and “Reset Scores” Features:

- **Task:** Create buttons that allow players to reset the game board independently and to reset both the board and the move counter/timer.
- **Hints:**

- Add two buttons in your HTML: one for “Reset Game” (which re-shuffles the cards and clears the current board state) and one for “Reset Scores” (which resets both the move counter/timer and the game board).
- Write JavaScript functions to handle each reset action, including a confirmation prompt before resetting scores if desired.
- Ensure that resetting the board maintains any selected settings (like grid size) unless a full reset is confirmed.

5. **(Advanced) Add Sound Effects:**

- **Task:** Integrate sound effects for card flips, successful matches, and game completion.
- **Hints:**
 - Use HTML5 <audio> elements to incorporate sounds.
 - Trigger audio playback with JavaScript when a card is flipped, when a pair is matched, or when all pairs are found.
 - Optionally, provide a mute/unmute option for the user.

6. **(Advanced) Introduce Difficulty Levels:**

- **Task:** Allow players to select a difficulty level that changes the game challenge (e.g., by varying the number of cards or adding a time constraint).
- **Hints:**
 - Modify the HTML to include a selection menu for difficulty levels.
 - Adjust the JavaScript board-generation logic based on the chosen difficulty.
 - Update the scoreboard and timer functions to work with the selected difficulty settings.

Recommended Resources:

- **HTML:**
[MDN HTML Tutorials](#)
Offers a comprehensive introduction to HTML, covering both fundamentals and advanced topics.
- **CSS:**
[MDN CSS Tutorials](#)
Explore the basics of CSS as well as modern layout techniques and animations.
- **JavaScript:**
[MDN JavaScript Tutorials](#)